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Fintech Final Write-Up

Note: All code and images are available on GitHub (Solutions/ contains all code; write-up/tables/ contains all images and the LaTeX source):
https://github.com/Kidus-Bezuayeho/fintech-final

Question 0

Part (a)

For an OLS regression to yield a causal estimate, it would need to be true that there is no omitted variable bias, no selection bias, and no reverse causality. This is because omitted variable bias can lead to a false relationship when a third factor drives both variables, selection bias can cause results to reflect the type of stocks that attract herding rather than herding itself, and reverse causality can make buying appear to predict returns when investors are actually reacting to expected returns.

Part (b)

One confounder is media and news coverage. When a sector gets a lot of media attention, Robinhood users rush to buy stocks in that sector while prices rise across the board, even for companies that are not actually benefiting. When prices eventually fall back down, it looks like the buying caused the drop when really the media hype was responsible all along.

Another confounder is earnings announcements. Before a company reports its earnings, speculation drives increased buying and price run-ups. If the report disappoints, prices fall, making the buying appear to have caused the drop when the earnings miss was the real cause.

Part (c)

The authors attempt to address identification concerns in several ways. One approach is controlling for lagged returns, which minimizes the effects of stocks naturally bouncing back after a big move. If a stock shoots up one day, it often falls the next simply because of normal price behavior, not because of any buying activity. By controlling for lagged returns, the authors can show that the negative returns they find are not just this normal bounce-back effect.

Another way they attempted to address this concern was with the outage analysis so that they could isolate the specific impact of Robinhood trading itself. On days when Robinhood went down, users who wanted to buy certain stocks simply could not. By comparing trading activity on outage days to normal days, the authors could show that the drop in trading was directly tied to Robinhood users being locked out, not to any other factor, making it much harder to argue that a third variable was driving the results.

Question 1

Part (a)

Please see code, Solutions/Question_1.py for how I calculated 1a, left comments in the code. For user change, I had to calculate it with a 3 day gap between days. Since the data set was a randomized subset of the full data, many data points did not have the previous day avaibale. I choose the 3 day gap as it would allow me observe Friday to MOnday changes while excluding gaps that were to far to be meaningful.

Part (b)

Panel A:

	N	Mean	SD	Min	p25	p50	p75	Max
users_close	1888	1764.6261	9411.3843	0.0	47.7500	185.5	747.0000	199222.0000
userchg	1888	6.8803	109.6439	-540.0	-1.0000	0.0	1.0000	3334.0000
userratio	1888	1.0139	0.2061	0.0	0.9984	1.0	1.0056	8.3778

Part (c)

```
Panel B:
  Mean # unique tickers per day:              74.4874
  Mean total user positions (summed/day):     153269.5186
  Median total user positions (summed/day):   94421.0000
```

Part (d)

When I compare my summary statistics to the paper, my subsample's distribution is comparable, though the scale differs due to the dataset size. In Panel A, the tendencies in my data are similar to the authors' findings. My median user count of 185.5 and average user ratio of 1.01 mirror the study's median of 160 and mean user ratio of 1.01. Both datasets show a positive skew which confirms that Robinhood user attention is focused on a few stocks. However, the maximums and daily aggregates are smaller in my subsample. The maximum daily user change in my sample is 3,334, compared to 85,193 in the paper. This makes sense because a random sample is less likely to have the massive trading events included. Additionally, my Panel B daily totals are lower. My sample averages 74 unique tickers and 153,000 total positions per day, whereas the full study tracks 7,211 tickers and 15 million daily positions.

Question 2

Part (a)

```
==================================================
  (a) rh_herd summary
==================================================
  Total herding events (rh_herd=1): 366
==================================================
```

Part (b)

There was a large discrepancy between the number of herding events in my sample and the paper's full dataset. My sample averaged 0.4851 herding events per day while the paper reported 9. I investigated why this difference existed and found it stems from the gaps between observations described in Question 1. The average gap between consecutive observations per ticker was ($\sim$)82 days. Because I cap the lookback at 3 days, most stocks have no valid userratio on any given day, leaving only ($\sim$)2.5 qualifying stocks per day. With such a small pool, the top 0.5% threshold flags roughly 1 out of every 2–3 stocks, making the indicator far less selective than it would be in the full dataset where ($\sim$)74 stocks qualify each day.

If I do not use the 3-day gap and instead use the closest prior observation regardless of how long ago it was, the mean and standard deviation of user change become far larger and noisier, no longer comparable to the paper. Therefore, I will stick with the 3-day gap despite it greatly reducing the number of qualifying stocks and herding events per day.

```
==================================================
  (b) Herding events vs. paper benchmark
==================================================
  Total herding events:    366
  Unique days in sample:   756
  Avg herding events/day:  0.4841
  Paper benchmark:         ~9 events/day
  Difference:              8.5159 fewer per day
==================================================
```

```
==================================================
  Stocks per day with prev observation <= 3 days
==================================================
  Average: 2.56
  Median:  2
==================================================

==================================================
  Avg gap between observations per ticker
==================================================
  82.71 days
==================================================


================================================================================
  userchg — no gap filter vs. 3-day gap filter
================================================================================
                        N      Mean         SD       Min  p25  p50   p75       Max
No gap filter     53880.0  449.4430  6830.6756 -163507.0 -1.0  3.0  44.0  634238.0
3-day gap filter   1940.0    6.6964   108.1694    -540.0 -1.0  0.0   1.0    3334.0
================================================================================
```

Part (c)

There was an issue where results from my sample for returns was lower than teh 366 herding events. I then discoverred that the issue was that many of the stocks seem to be delisted from yfinance. I included one example where a stock called ACT was mentioned in the dataset (I only included 5 mentions of it) but when I attempted to pull from yfinance you can see it return that it is not returning anything. Please see Solutions/debug_ACT.py to see how I tested this issue.

```
==========================================================================================
  (c) Summary statistics — herding event subsample (rh_herd = 1)
==========================================================================================
                     N       Mean          SD       Min       p25       p50        p75          Max
users_close      366.0  3358.4180  14144.9901  101.0000  225.0000  490.0000  1587.7500  183168.0000
userchg          366.0    44.0055    212.5609    1.0000    2.0000    5.0000    17.0000    3334.0000
userratio        366.0     1.0393      0.2292    1.0000    1.0052    1.0110     1.0229       5.0803
returns          188.0    -0.0010      0.0361   -0.1552   -0.0120    0.0008     0.0104       0.2083
abnormal_returns 188.0    -0.0011      0.0336   -0.1557   -0.0111    0.0007     0.0099       0.2034
==========================================================================================
```

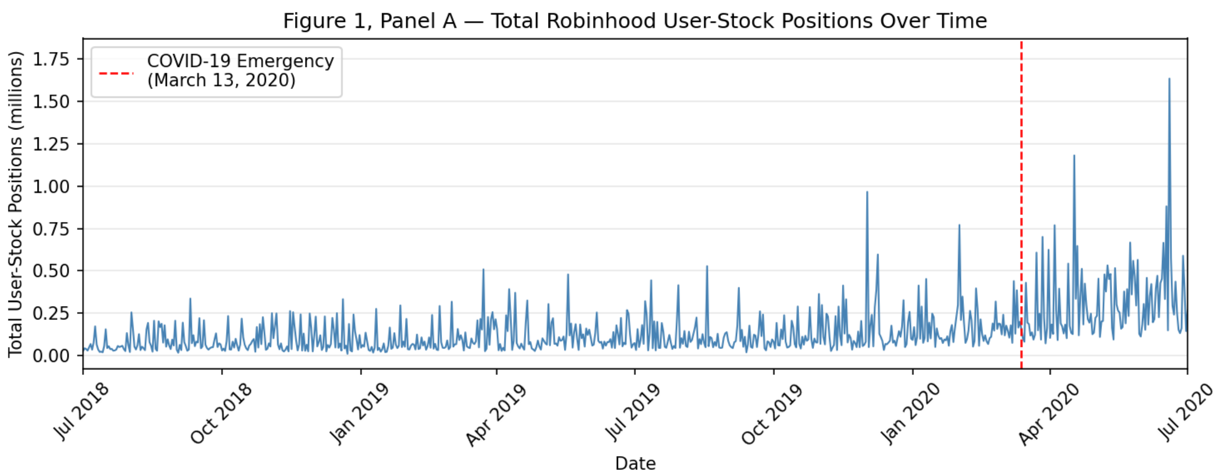
```
============================================================
  ACT in Robintrack_subsample.csv
============================================================
Ticker            DateTime  users_lag  Users  minutes  delay   random       Date
   ACT 2020-05-07 17:00:00      981.0  981.0     42.0    0.0 0.000039 2020-05-07
   ACT 2020-07-18 12:00:00      924.0  924.0     44.0    0.0 0.000069 2020-07-18
   ACT 2019-08-22 08:00:00     1108.0 1108.0     41.0    0.0 0.000089 2019-08-22
   ACT 2020-06-19 14:00:00      952.0  951.0     42.0    0.0 0.000145 2020-06-19
   ACT 2019-11-02 20:00:00     1059.0 1059.0     41.0    0.0 0.000167 2019-11-02
============================================================

============================================================
  yfinance download attempt for ACT
============================================================
$ACT: possibly delisted; no price data found  (1d 2018-01-01 -> 2021-01-01) (Yahoo error = "Data doesn't exist for startDate = 1514782800,
 endDate = 1609477200")

1 Failed download:
['ACT']: possibly delisted; no price data found  (1d 2018-01-01 -> 2021-01-01) (Yahoo error = "Data doesn't exist for startDate = 15147828
00, endDate = 1609477200")
  Result: no data returned (ticker likely delisted)
============================================================
```

Part (d)

The graph does show a similar story to the graphs of the study, we can see that after COVID the peaks of the users positions seem to be much higher than before COVID. However, due to the my smaller sample size, my graph has much more volaitilty than the paper's graph. This is most likely why it is not showing the same strong upward trend that the paper's graph.



Question 3

Part (a)

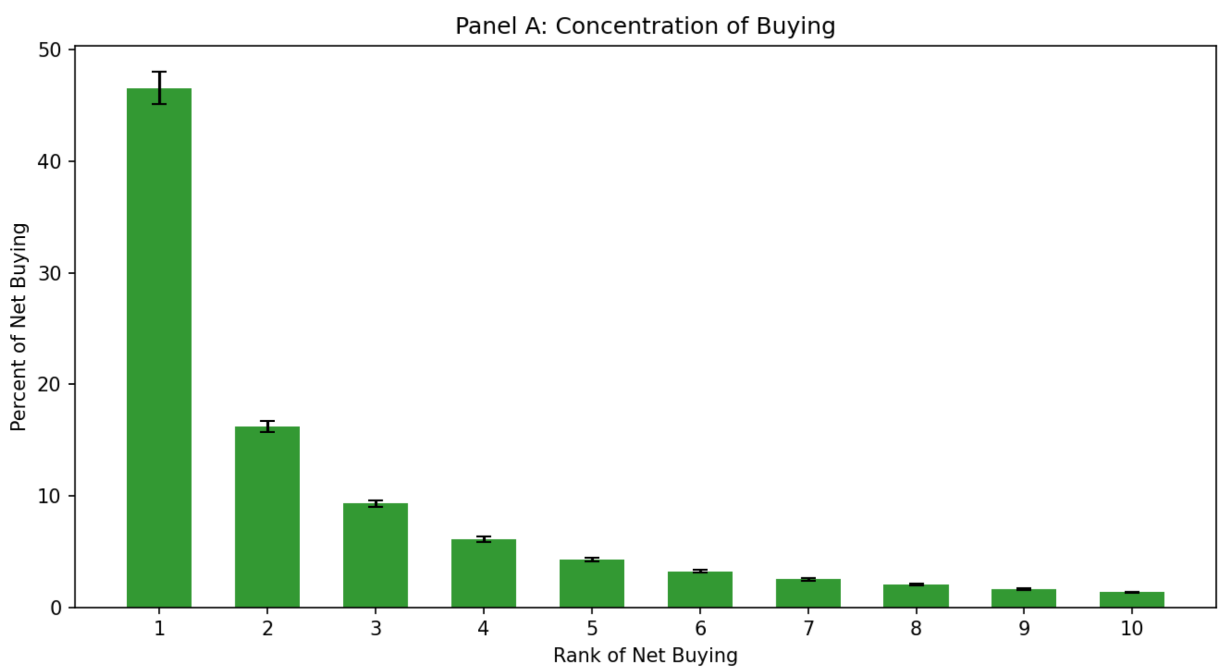
For this problem, I calculated user change without the 3-day gap. This is because with the 3-day each day would only have 6 stocks with valid values. Since that would just include all 6 for each day, I opted to use the closest prior observation regardless of how long ago it was so that there would be more stocks per day to consider.

```
==================================================================
  (a) Top-10 buying concentration (Top10_buy)
==================================================================
N        830.0000
Mean       0.9256
SD         0.0626
Min        0.6462
p25        0.8922
p50        0.9389
p75        0.9743
Max        1.0000
==================================================================
```

Part (b)

```
==================================================================
  (b) Bottom-10 selling concentration (Top10_sell)
==================================================================
N        828.0000
Mean       0.9613
SD         0.0415
Min        0.7649
p25        0.9427
p50        0.9738
p75        0.9951
Max        1.0000
==================================================================
```

Part (c)



Part (d)

In this comparison, we can see that the overall shape of the data is identical to that of the study, with rank one having the majority of the concentration, a sharp drop from rank 1 to rank 2, then a slow decline from rank 2 to 10. However, the scale of my graph is much larger than that of the paper. My graph shows that the top 10 stocks have a concentration of about 93% of the buying while the paper's is 35%. This difference is likely due to my subsample having far fewer stocks observed per day, so the top 10 captures nearly all active stocks by construction. Large gaps between observations can also make the increase in users appear more extreme than it really is, which could further inflate concentration in the top 10 stocks.

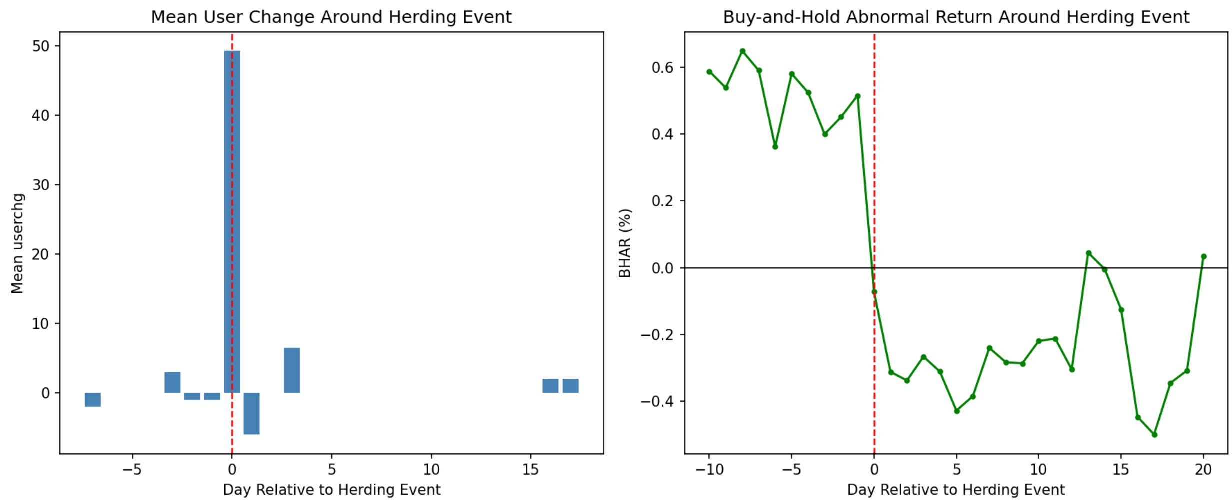
Question 4

Part (a)

BHAR is a better estimator of returns in this situation because it accurately captures the highly volatile nature of these stocks. When a stock increases in price from A to B, and then drops back down to A, the percentage increase on the way up is mathematically larger than the percentage decrease on the way down. This is because the peak price (B) is larger than the starting price (A), meaning the drop from B to A represents a smaller proportion of the price. Because of this, if you simply add the daily returns together (CAR), it will incorrectly say that you have a positive return, despite you being in the exact same net position with no real gains. BHAR prevents this phantom positive return.

Economically, this makes even more sense. When users see stocks driving upward in the Top Movers section on Robinhood, they invest in the same stock trying to catch the upward gains. However, all this does is artificially drive up the price; the actual value of the company did not really change. Intelligent investors will notice this, and as the investing frenzy cools, they will sell their shares while those who are less informed continue to hold. This causes the individuals who bought during the upward drive to lose money, as the stock will eventually go back down to its appropriate price.

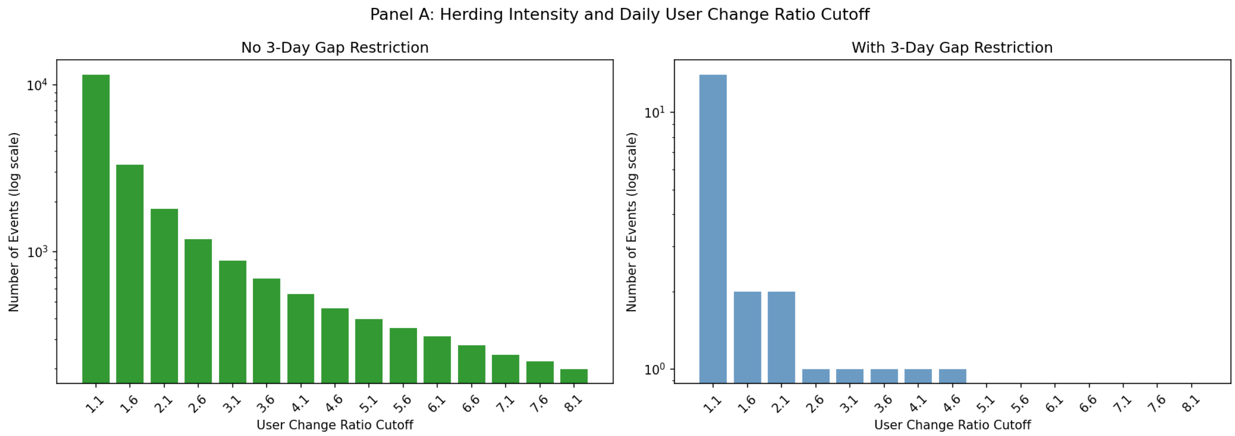
Part (b)



(**Note:** I used the 3-day gap restriction for this problem as well. I attempted to use no restrictions and the graphs were not lining up with the paper. The 3-day gap keeps the data cleaner by ensuring observations are close enough in time that user change values are meaningful. To see what the graph looked like without restrictions, please look at github repo: write-up/tables/4b_without_restrictions.png)

The pattern noticed in my graphs agrees with the paper. I see that the vast majority of user change is seen on Day 0, and afterwards there is a drop in effect. There are large gaps in my mean userchg graph due to my restriction of 3-days, but it still tells the most important part of the data. The BHAR also shows a similar trend where after Day 0 there is a downward trend on returns due to the buying hype dying down. When the massive wave of users stops buying the stock, the artificial demand goes away and the price drops back to what the company is actually worth.

Part (c)



This pattern is consistent with what we see in the paper, as User Change ratio increases, the number of events also decreases. This is because the higher the user change ratio, the more the extreme the event is which make them much more unlikely to occur. The paper uses line graphs and has much more observations, but the overall shape and trend of my graphs are identical to theirs.

Additionally, the pattern we see in these graphs matches the return reversals in the paper. As we move down to the right side of each graph, we are able to highlight the most extreme herding events. Since these are the most extreme events, their prices are driven extremely high and therefore have a much bigger crash to return to their true value, which proves the temporary price pressure mechanism.

(**Note:** I included two graphs, one with rescrtions and one without. They both told a simialr story so I included both to show that the pattern is consistent regardless of restrictions.)

Question 5

1. What does this finding imply about the informational efficiency of the market regarding Robinhood herding?

    This implies that the market is not fully efficient. Investors with more information understand that the price increase is due to herding and therefore will eventually come back down. Therefore, they create short positions to profit from the expected drop in price in order to turn a profit. If markets were efficient, there would be no herding and stocks would be priced at their true value at all times.

2. Does the presence of short sellers fully prevent the price impact of herding? Why or why not?

    The presence of short sellers does not fully prevent the price impact of herding. There are a couple reasons for this. First, the price of short selling can be high depending on the stock which can limit the number of short sellers willing to take the trade. Secondly, the timing of the short is important, if a short seller enters too early, they can have their position squeezed by the herding and eventually force the position to close. Due to this, short sellers are unable to fully offset the herding effect.

3. How does this relate to the broader debate about whether fintech democratization helps or harms retail investors?

    This relates to the broader debate because the average retail investor is not going to be able to compete with informed investors and instituions who are able to short sell and profit from the herding. This means that the democratization of fintech can harm many retail investors who attempt to regularly trade stocks. Overall, retail investors often end up on the losing side of these trades. While fintech makes it easier to enter the market, it also makes it easier for regular people to get caught up in bubbles.